

Machine Learning for Wind Farm Output Forecasting

Highlights and Limitations

42578 Advanced Business Analytics

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Highlights

1. Weather prediction: In the first iterations of our modeling, using Facebook's Prophet in a univariate way we were not able to get good enough predictions. So, firstly we predicted 5 variables that influence wind speed (Granger causality) and combined them to predict the wind speed in a multivariate way. Still, this poses a chicken and egg problem, should we predict all variables univariate first and then predict all of them again in a multivariate setting with these predictions?
2. Web scraping for benchmarking: By using web scrapers/crawlers we were able to grab the data from The Wind Power website, which does not provide direct access to it on a large scale. Afterwards, we used this data in our case study to compare how well our models predict. On most websites, this data is locked or not even publicly available, so we were lucky this website posts a preview of the real data every day.
3. Trying different kinds of models and interpreting them: For the capacity factor prediction, we tried three different types of neural networks, a random forest and a Support Vector Machine, as well as an XGBoost model, in order to find the most suitable one. Furthermore, for the XGBoost model, which we selected in the end, we also did an interpretability study using the SHAP library.
4. Combining several models for case studies: What also makes our project valuable is that for our case studies, we combined several different models: Firstly, multiple weather prediction models, and secondly, capacity factor prediction models which use the output of the weather prediction as an input. Furthermore, we applied different filtering methods based on literature research to identify possible new locations. On these filtered locations, we applied a similarity measure which gave us the most similar location among those used for our initial modeling. Based on this classification, we then applied our models, in order to select the most promising new wind farm locations.

Limitations

1. As always, the data quality. Wind farms do not have the data on their location, so we grabbed the data from the closest weather stations. In some cases, the distance is over 30km, which is definitely not ideal. Additionally, a lot of variables have NaN or empty values; or, have data that is limited to 2 or 3 years which often does not suffice to create good models. Furthermore, it is very difficult to find real energy generation data for wind farms. This is why we used simulated data for the capacity factors on a grid of locations in Europe.
2. Computational power. Facebook's Prophet library is very power-hungry in general even when not running cross-validation (CV). And, in order to tune the model with hyperparameters

you need to run CV, and without it tuning it manually is just a terrible experience. We did manual tuning with the all of the models, so they work very well, but is it not feasible for a large amount of windfarms. For future work, having more power would definitely make our lives easier especially with the amount of models we have per wind farm.

3. Due to the limited experience in the onshore wind farm sector, the project has analyzed a limited number of meteorological predictor variables. Additionally, the wind turbine model, blade length, turbine position or even terrain topology could attenuate the wind speed and affect the capacity factor. However, there may be hidden predictor variables that directly affect wind energy production that have not been considered due to this limitation. Furthermore, the use of powerful and complex time series analysis tools, which are not always open source, can also pose a challenge in predicting weather patterns and optimizing wind energy production.
4. The outcome we are trying to achieve is based on multiple predictions all of which are characterized by an error due to the above limitations. By applying them consecutively the error 'sums up' and that is why we cannot be as accurate as we would like to be in our predictions. If in the future we find a way to resolve some of the above limitations then the overall accuracy would increase.